Program Schemes for Bachelor of Engineering (Hons.) (Computer Science and Engineering) (with Specialization in Artificial Intelligence and Machine Learning)

Cluster: Engineering & Technology

Institute: University Institute of Engineering

Department: AIT-CSE

Type of Program: UG

SEMESTER-1		
Course Name	**Course Category**	**Credits**
LOGICAL THINKING AND PROBLEM SOLVING	Major Core	4
PROFESSIONAL CERTIFICATION-I	Skill Enhancement	1
DESIGN THINKING AND INNOVATION	Major Core	1
ACADEMIC AND CAREER PATHWAY PLANNING	Value Education	1
DIGITAL ELECTRONICS	Major Core	4
AI APPLICATIONS LAB	Major Core	1
COMMUNICATION SKILLS -I	Ability Enhancement	1
MATHEMATICS-I	Major Core	5
BIOLOGY FOR ENGINEERS	Major Core	3
UNIVERSAL HUMAN VALUES, ETHICS AND LIFE SKILLS-1	University Core	1

SEMESTER-2		
Course Name	**Course Category**	**Credits**
DATA STRUCTURES & ALGORITHMS	Major Core	4
MATHEMATICS-II	Major Core	5
SEMICONDUCTOR PHYSICS AND APPLICATION	Major Core	3
COMMUNICATIONS SKILLS - II	Ability Enhancement	1
EMERGING & DISRUPTIVE TECHNOLOGIES WORKSHOP	Major Core	1
PROGRAMMING PRACTICE	Major Core	1
SOFT SKILLS-I	Skill Enhancement	1
GENERAL PROFICIENCY-I	Ability Enhancement	1
ENGINEERING PHYSICS	Major Core	4
COMPUTER GRAPHICS LAB	Major Core	1

SEMESTER-3		
Course Name	**Course Category**	**Credits**
COMPUTER ORGANIZATION AND ARCHITECTURE	Major Core	3
ADVANCED DATA STRUCTURES	Major Core	4
DATABASE MANAGEMENT SYSTEM	Major Core	4
AGILE DEVELOPMENT METHODOLOGIES	Major Core	4
OBJECT ORIENTED PROGRAMMING USING JAVA	Major Core	2
PROBABILITY AND STATISTICS	Major Core	3
UNIVERSAL HUMAN VALUES, ETHICS AND LIFE SKILLS-2	Value Education	2
SOCIAL INTERNSHIP	Summer Internship	2
APTITUDE-I	Ability Enhancement	1
SEMESTER MINI PROJECT	Project	2
ENTREPRENEURSHIP MINDSET	Ability Enhancement	1
CRITICAL ENQUIRY AND RESEARCH METHODOLOGY	Ability Enhancement	1
GENERAL PROFICIENCY-II	Ability Enhancement	1

SEMESTER-4		
Course Name	**Course Category**	**Credits**
OPERATING SYSTEM	Major Core	4
DESIGN AND ANALYSIS OF ALGORITHMS	Major Core	4
SYSTEM DESIGN	Major Core	3
PROFESSIONAL ELECTIVE -I	Program Elective	4
STATISTICAL METHODS FOR AI	Major Core	4
PYTHON FOR MACHINE LEARNING	Major Core	3
RIE-1	Ability Enhancement	1
RIE-2	Skill Enhancement	1
GENERAL PROFICIENCY-III	Skill Enhancement	1
COMPETITIVE CODING – I	Skill Enhancement	1

SEMESTER-5		
Course Name	**Course Category**	**Credits**
FOUNDATIONS OF MACHINE LEARNING	Major Core	4
FULL STACK DEVELOPMENT - I	Skill Enhancement	2
COMPUTER NETWORKS	Major Core	4
NEURAL NETWORK DESIGN AND OPTIMIZATION	Major Core	4
COMPETITIVE CODING – II	Skill Enhancement	1
PROFESSIONAL ELECTIVE -II	Program Elective	1
PROFESSIONAL ELECTIVE -III	Program Elective	3
TECHNICAL INTERNSHIP	Internship	1
RIE-3	Skill Enhancement	1
RIE-4	Ability Enhancement	1
GENERAL PROFICIENCY-IV	Skill Enhancement	1

SEMESTER-6		
Course Name	**Course Category**	**Credits**
FULL STACK DEVELOPMENT - II	Skill Enhancement	2
REINFORCEMENT LEARNING AND XAI	Program Elective	4
ADVANCED DATABASE MANAGEMENT SYSTEM	Skill Enhancement	1
THEORY OF COMPUTATION	Major Core	3
PROFESSIONAL ELECTIVE-IV	Program Elective	3
PROFESSIONAL ELECTIVE-V	Program Elective	3
CAPSTONE PROJECT – I	Project	3
RIE-5	Skill Enhancement	1
RIE-6	Ability Enhancement	1
COMPETITIVE CODING – III	Skill Enhancement	1
SOFTWARE ENGINEERING	Major Core	2

SEMESTER-7		
Course Name	**Course Category**	**Credits**
DEEP LEARNING	Major Core	4
PROFESSIONAL ELECTIVE-VI	Program Elective	3
CAPSTONE PROJECT – II	Project	3
OPEN ELECTIVE-I	Open Elective	3
ENVIRONMENTAL SCIENCE, WASTE AND DISASTER MANAGEMENT	University Core	2
INDUSTRIAL INTERNSHIP	Internship	1
COMPETITIVE CODING – IV	Skill Enhancement	1

SEMESTER-8		
Course Name	**Course Category**	**Credits**
INTELLIGENT AGENTS AND AGENTIC AI	Program Elective	4
RESEARCH PROJECTS & PAPER	Project	2
CAPSTONE AI PROJECT – III	Project	5
OPEN ELECTIVE-II	Open Elective	3
OPEN ELECTIVE-III	Open Elective	3
PROFESSIONAL ETHICS	University Core	2

Program Electives			
Semester	**Course Name**	**Course Category**	**Credits**
3rd Sem	FOUNDATION OF DEVOPS	Program Elective	4
3rd Sem	LINUX PROGRAMMING	Program Elective	2
4th Sem	DATA PREPROCESSING AND FEATURE ENGINEERING	Program Elective	4
4th Sem	INTRODUCTION TO DATA SCIENCE	Program Elective	4
4th Sem	DATA MINING AND DATA WAREHOUSE	Program Elective	4
5th Sem	FOUNDATIONS OF MACHINE LEARNING	Program Elective	4
5th Sem	SOFT COMPUTING	Program Elective	4

5th Sem	BIG DATA & HADOOP	Program Elective	3
5th Sem	FUNDAMENTALS OF QUANTUM COMPUTING	Program Elective	3
5th Sem	PREDICTIVE ANALYTICS MODELLING	Program Elective	3
5th Sem	PROGRAMMING IN R	Program Elective	1
5th Sem	DOCKER AND KUBERNETES	Program Elective	1
5th Sem	EXPLORATORY DATA ANALYSIS	Program Elective	1
6th Sem	MLOPS FOR DATA SCIENTISTS	Program Elective	3
6th Sem	NATURAL LANGUAGE PROCESSING	Program Elective	3
6th Sem	ARTIFICIAL INTELLIGENCE	Program Elective	3
6th Sem	EDGE AI AND TINY MACHINE LEARNING	Program Elective	3
6th Sem	COMPUTER VISION	Program Elective	3
7th Sem	LARGE LANGUAGE MODELS AND GENERATIVE AI	Program Elective	3
7th Sem	QUANTUM MACHINE LEARNING	Program Elective	3
7th Sem	SECURITY AND DATA PRIVACY LAWS AND STANDARDS	Program Elective	3
7th Sem	PROMPT ENGINEERING	Program Elective	3
7th Sem	UNIVERSITY OPEN ELECTIVE	University Core	3
7th Sem	UNIVERSITY OPEN ELECTIVE	University Core	3
8th Sem	UNIVERSITY OPEN ELECTIVE	University Core	3